1. Contents and safety

Contents

1. Contents and safety 1
2. Specifications 2
3. Control panel 3
4. Coolant circuit and valves 4
5. Electrical connections 5
6. Fault codes 6
7. Maintenance schedule 7
8. Strainer basket replacement 8

Safety

Read this manual before installing, operating or servicing the CR-220.

Lock out and tag out the main disconnect before opening any panel. The fan and pump can start without warning when the controller calls for cooling.

Only a certified refrigeration technician may open the refrigerant circuit.

The CR-220 holds R-513A refrigerant under pressure; wear eye protection and gloves. Never vent refrigerant to the atmosphere.

Coolant may be hot after operation. Let the reservoir cool below 40 C before draining it.

Warranty

Brindlemoor warrants the CR-220 against defects in materials and workmanship for 24 months from the date of shipment. Contact your Brindlemoor dealer.

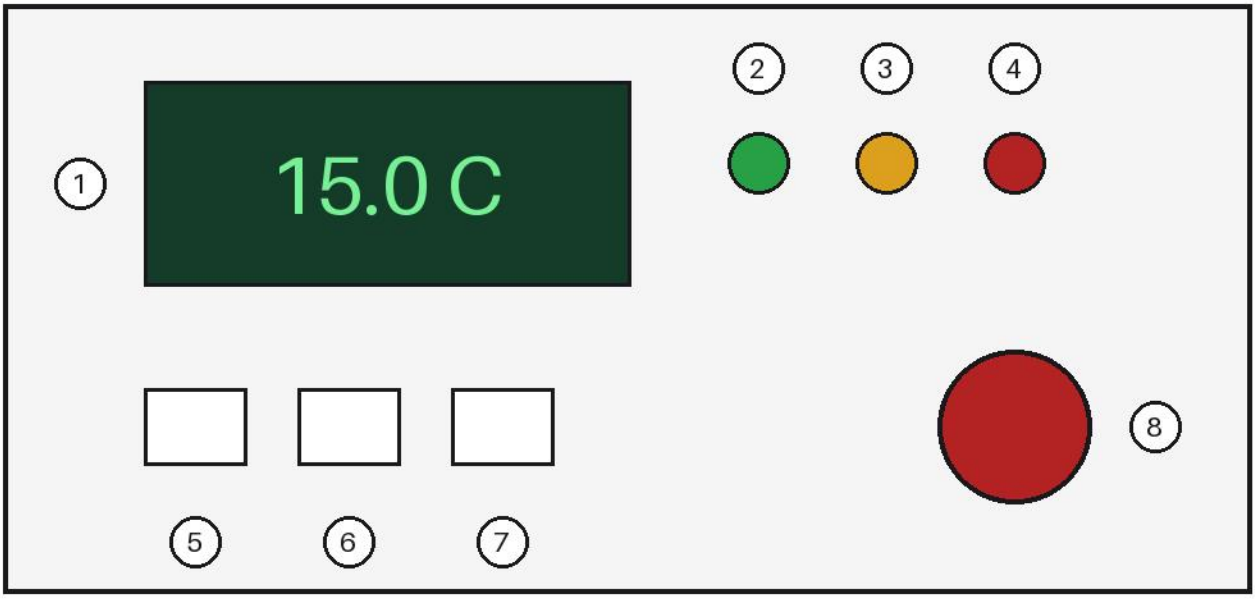
2. Specifications

Item	Value
Cooling capacity	22 kW at 15 C leaving coolant, 35 C ambient
Refrigerant	R-513A
Refrigerant charge	4.6 kg
Power supply	400 V, 3 phase, 50 Hz
Full load current	18.5 A
Coolant	Water or up to 30 % propylene glycol
Reservoir volume	60 L
Nominal coolant flow	65 L/min
Pump maximum pressure	3.5 bar
Setpoint range	5 C to 30 C
Sound level	62 dB(A) at 1 m
Weight, empty	210 kg
Dimensions (W x D x H)	760 x 1150 x 1420 mm

Ambient operating range: 5 C to 43 C. Above 43 C ambient the controller reduces capacity and shows fault E04.

3. Control panel

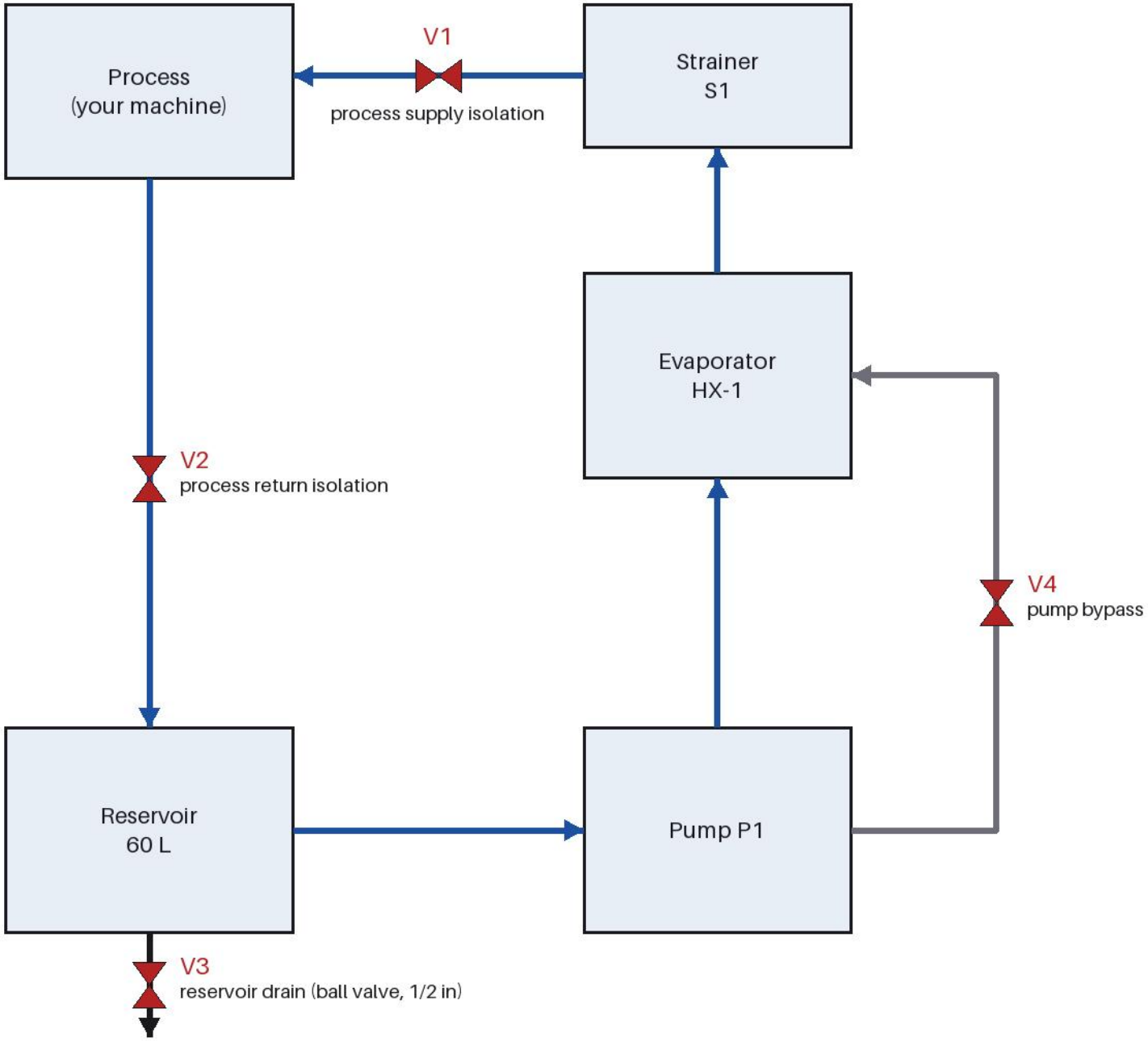
Figure 3-1: front control panel. Numbers refer to the list below.



1 Display: leaving coolant temperature, or the fault code when a fault is active
2 Green lamp: running
3 Amber lamp: warning, the chiller keeps running
4 Red lamp: fault, the compressor is stopped
5 SET: hold 3 s to change the setpoint
6 UP / DOWN: change the value shown
7 RESET: clears a fault once its cause is gone
8 Emergency stop: stops the pump, fan and compressor at once

4. Coolant circuit and valves

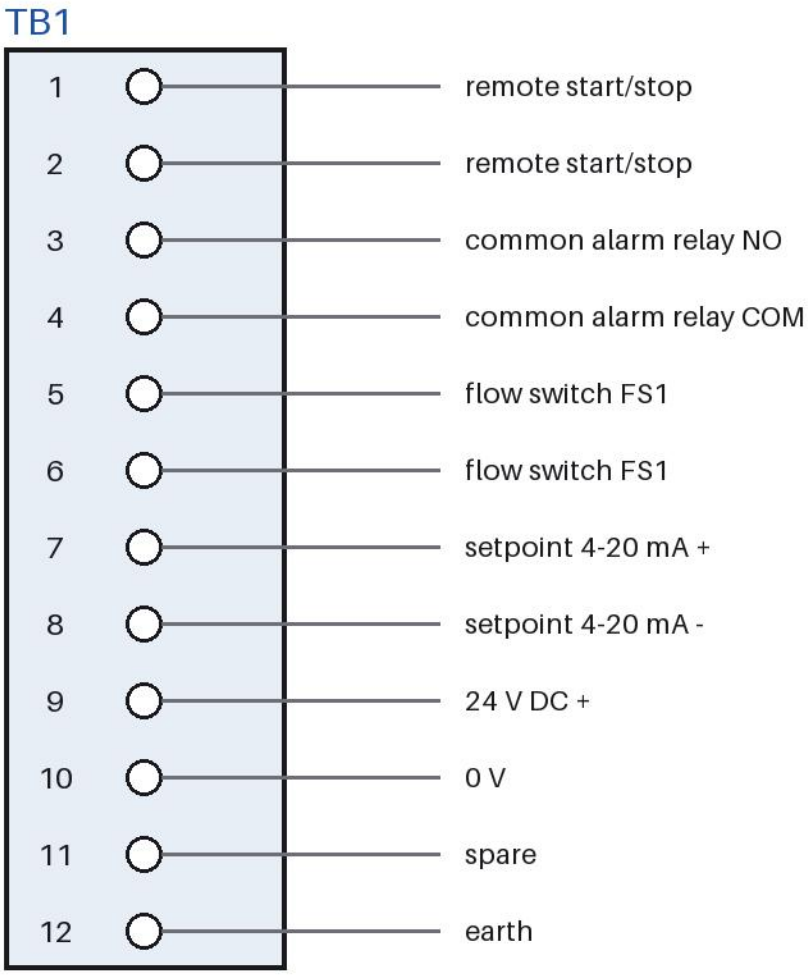
Figure 4-1: coolant circuit. Arrows show the direction of flow.



Close V1 and V2 to isolate the chiller from the process without draining it. Keep V4 closed in normal operation.

5. Electrical connections

Figure 5-1: customer terminal block TB1, inside the electrical panel door. Use 0.75 to 1.5 mm2 wire. Low-voltage signals only on TB1.



A closed dry contact across the remote start/stop pair runs the chiller; open it to stop. Remove the factory jumper before connecting a remote switch.

6. Fault codes

The display shows the active fault code. Press RESET once the cause is fixed.

Code	Meaning	What to do
E01	High refrigerant pressure	Clean condenser coil; check fan
E02	Low refrigerant pressure	Call a refrigeration technician
E03	Compressor overload	Check supply voltage; call service
E04	Ambient temperature high	Improve ventilation around unit
E05	Leaving coolant too cold	Check setpoint; check glycol mix
E06	Leaving coolant too warm	Check process heat load
E07	Low coolant flow	Open V1 and V2; clean strainer S1
E08	Low reservoir level	Top up coolant to the MAX mark
E09	Temperature sensor fault	Check sensor plug T1; call service
E10	Phase loss or reversal	Check 400 V supply phases

Faults E02, E03 and E09 need a service visit if they return after a reset.

7. Maintenance schedule

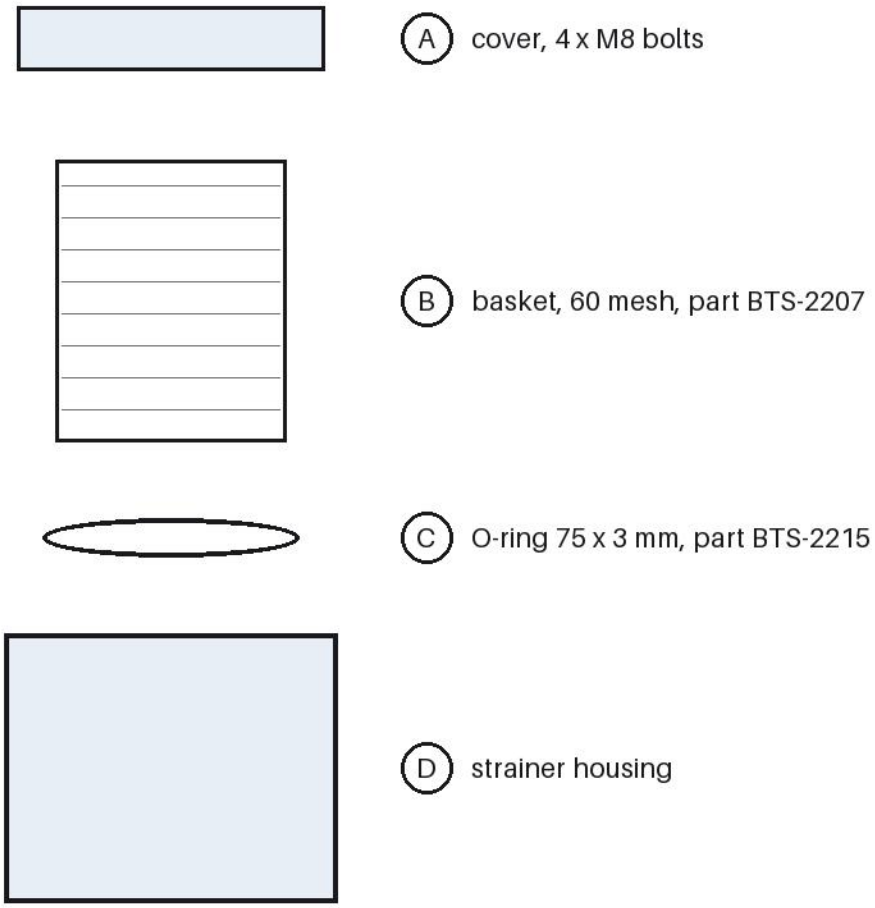
Lock out the main disconnect before any task marked with an asterisk (*).

Interval	Task
Daily	Check the reservoir level on the sight glass
Daily	Check for leaks around the pump and fittings
Weekly	Clean the strainer basket S1 * (see page 8)
Monthly	Clean the condenser coil with compressed air *
Monthly	Check glycol concentration with a refractometer
Every 6 months	Check pump seal for weeping *
Every 6 months	Tighten power terminals *
Yearly	Drain and replace the coolant *
Yearly	Refrigerant leak check by a certified technician

Record each task, the date and your initials in the maintenance log at the back of the electrical panel door.

8. Strainer basket replacement

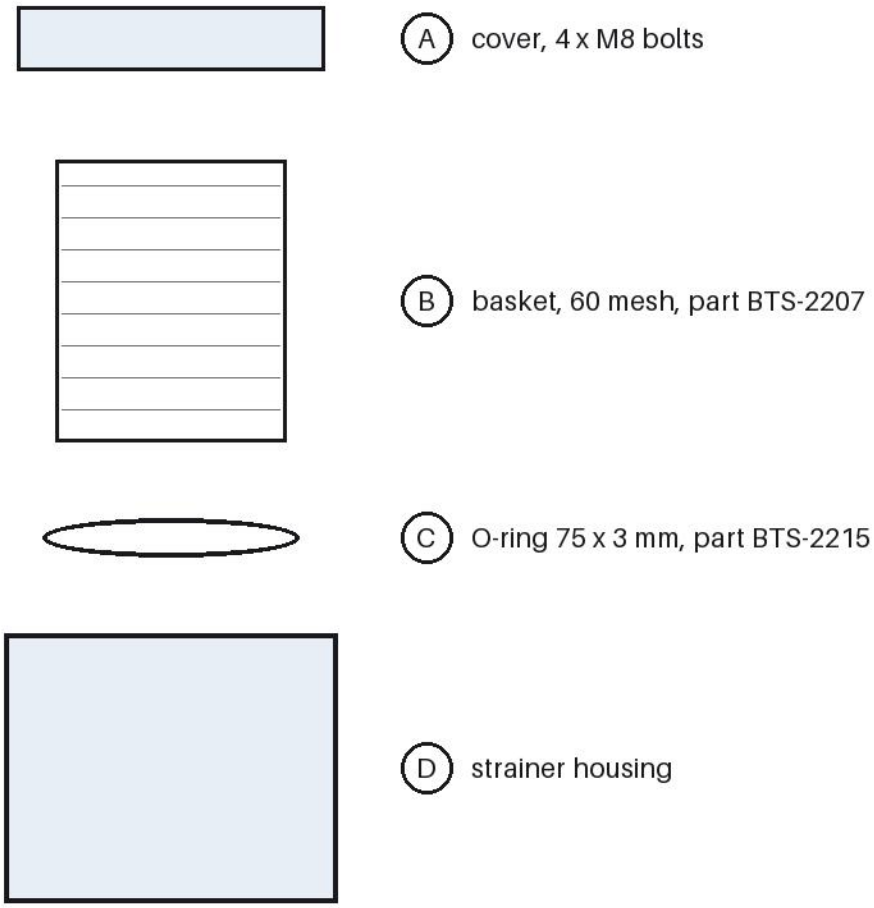
Figure 8-1: strainer S1, exploded view.



1. Stop the chiller and lock out the main disconnect.
2. Close V1 and V2 (page 4) and let the pressure fall to zero.
3. Remove the four cover bolts and lift off the cover.
4. Pull out the basket. Rinse it, or fit a new basket BTS-2207.
5. Fit a new O-ring BTS-2215 every time the cover is opened.
6. Refit the cover and tighten the bolts to 20 N m in a cross pattern.
7. Open V1 and V2, start the chiller and check for leaks.

8. Strainer basket replacement

Figure 8-1: strainer S1, exploded view.



1. Stop the chiller and lock out the main disconnect.
2. Close V1 and V2 (page 4) and let the pressure fall to zero.
3. Remove the four cover bolts and lift off the cover.
4. Pull out the basket. Rinse it, or fit a new basket BTS-2207.
5. Fit a new O-ring BTS-2215 every time the cover is opened.
6. Refit the cover and tighten the bolts to 25 N m in a cross pattern.
7. Open V1 and V2, start the chiller and check for leaks.